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GATE CSE · DATABASE MANAGEMENT SYSTEM · CHAPTER-WISE PYQS

Functional Dependencies and Normalization

GATE CSE Database Management System · Previous Year Questions with Answer Key

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Answer key	at the end

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Marks 1 22 questions

Q1. GATE CSE 2026 Set 1

Let P, Q, R and S be the attributes of a relation in a relational schema. Let $X \rightarrow Y$ indicate functional dependency in the context of a relational database, where $X, Y \subseteq \{P, Q, R, S\}$ Which of the following options is/are always true?

(A) If $(\{P, Q\} \rightarrow \{R\}$ and $\{P\} \rightarrow \{R\})$, then $\{Q\} \rightarrow \{R\}$

(B) If $\{P, Q\} \rightarrow \{R\}$, then $(\{P\} \rightarrow \{R\}$ or $\{Q\} \rightarrow \{R\})$

(C) If $(\{P\} \rightarrow \{R\}$ and $\{Q\} \rightarrow \{S\})$, then $\{P, Q\} \rightarrow \{R, S\}$

(D) If $\{P\} \rightarrow \{R\}$, then $\{P, Q\} \rightarrow \{R\}$

Open solution

Q2. GATE CSE 2026 Set 1

In the context of relational database normalization, which of the following statements is/ are true?

(A) It is always possible to obtain a dependency-preserving 3NF decomposition of a relation

(B) It is always possible to obtain a dependency-preserving 1NF decomposition of a relation

(C) It is not always possible to obtain a dependency-preserving BCNF decomposition of a relation

(D) It is not always possible to obtain a dependency-preserving 2NF decomposition of a relation

Open solution

Q3. GATE CSE 2024 Set 1

Which of the following statements about a relation R in first normal form (1NF) is/are TRUE?

(A) R can have a multi-attribute key

(B) R cannot have a foreign key

(C) R cannot have a composite attribute

(D) R cannot have more than one candidate key

Open solution

Q4. GATE CSE 2022

In a relational data model, which one of the following statements is TRUE?

(A) A relation with only two attributes is always in BCNF.

(B) If all attributes of a relation are prime attributes, then the relation is in BCNF.

(C) Every relation has at least one non-prime attribute.

(D) BCNF decompositions preserve functional dependencies.

Open solution

Q5. GATE CSE 2022

Consider a relation R(A, B, C, D, E) with the following three functional dependencies.

AB $\rightarrow$ C ; BC $\rightarrow$ D ; C $\rightarrow$ E;

The number of superkeys in the relation R is _____.

Open solution

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Q6. GATE CSE 2021 Set 2

Open solution

Consider the following statements S1 and S2 about the relational data model:

S1: A relation scheme can have at most one foreign key.

S2: A foreign key in a relation scheme R cannot be used to refer to tuples of R.

Which one of the following choices is correct?

(A) S1 is true and S2 is false.

(B) Both S1 and S2 are true.

(C) Both S1 and S2 are false.

(D) S1 is false and S2 is true.

Q7. GATE CSE 2017 Set 1

Open solution

The following functional dependencies hold true for the relational schema $R\{V, W, X, Y, Z\}$:

$V \rightarrow W$
$VW \rightarrow X$
$Y \rightarrow VX$
$Y \rightarrow Z$

Which of the following is irreducible equivalent for this set of functional dependencies?

(A) $V \rightarrow W$
$V \rightarrow X$
$Y \rightarrow V$
$Y \rightarrow Z$

(B) $V \rightarrow W$
$W \rightarrow X$
$Y \rightarrow V$
$Y \rightarrow Z$

(C) $V \rightarrow W$
$V \rightarrow X$
$Y \rightarrow V$
$Y \rightarrow X$
$Y \rightarrow Z$

(D) $V \rightarrow W$
$W \rightarrow X$
$Y \rightarrow V$
$Y \rightarrow X$
$Y \rightarrow Z$

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Q8. GATE CSE 2016 Set 1

Open solution

A database of research articles in a journal uses the following schema.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, YEAR, PRICE)

The primary key is **(VOLUME, NUMBER, STARTPAGE, ENDPAGE)** and the following functional dependencies exist in the schema.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE) → TITLE
(VOLUME, NUMBER) → YEAR
(VOLUME, NUMBER, STARTPAGE, ENDPAGE) → PRICE

The database is redesigned to use the following schemas.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, PRICE)
(VOLUME, NUMBER, YEAR)

Which is the weakest normal form that the new database satisfies, but the old one does not?

(A) 1NF (B) 2NF (C) 3NF (D) BCNF

Q9. GATE CSE 2015 Set 3

Open solution

Consider the relation $X\,(P, Q, R, S, T, U)$ with the following set of functional dependencies

$$F = \{ \\ \{P, R\} \rightarrow \{S, T\}, \\ \{P, S, U\} \rightarrow \{Q, R\} \\ \}$$

Which of the following is the trivial functional dependency in F^+, where F^+ is closure of F ?

(A) $\{P, R\} \rightarrow \{S, T\}$ (B) $\{P, R\} \rightarrow \{R, T\}$

(C) $\{P, S\} \rightarrow \{S\}$ (D) $\{P, S, U\} \rightarrow \{Q\}$

Q10. GATE CSE 2014 Set 1

Open solution

Consider the relation scheme $R = (E, F, G, H, I, J, K, L, M, N)$ and the set of functional dependencies $\{\{E, F\} \rightarrow \{G\}, \{F\} \rightarrow \{I, J\}, \{E, H\} \rightarrow \{K, L\}, \{K\} \rightarrow \{M\}, \{L\} \rightarrow \{N\}\}$ on R. What is the key for R?

(A) $\{E, F\}$ (B) $\{E, F, H\}$

(C) $\{E, F, H, K, L\}$ (D) $\{E\}$

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Q11. GATE CSE 2014 Set 2

Open solution

Given an instance of the STUDENTS relation as shown below:

StudentID	StudentName	StudentEmail	StudentAge	CPI
2345	Shankar	shankar@math	X	9.4
1287	Swati	swati@ee	19	9.5
7853	Shankar	shankar@cse	19	9.4
9876	Swati	swati@mech	18	9.3
8765	Ganesh	ganesh@civil	19	8.7

For ***(StudentName, StudentAge)*** to be a key for this instance, the value X should NOT be equal to _____.

Q12. GATE CSE 2014 Set 2

Open solution

The maximum number of superkeys for the relation schema $R(E, F, G, H)$ with E as key is _____.

Q13. GATE CSE 2014 Set 3

Open solution

A prime attribute of a relation scheme R is an attribute that appears

(A) In all candidate keys of R

(B) In some candidate key of R.

(C) In a foreign key of R.

(D) Only in the primary key of R.

Q14. GATE CSE 2012

Open solution

Which of the following is TRUE?

(A) Every relation in $3NF$ is also in $BCNF$

(B) A relation R is in $3NF$ if every non-prime attribute of R is fully functionally dependent on every key of R

(C) Every relation in $BCNF$ is also in $3NF$

(D) No relation can be in both $BCNF$ and $3NF$

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Q15. GATE CSE 2011

Consider a relational table with a single record for each registered student with the following attributes.

1. *Registration_Num*: Unique registration number of each registered student
2. *UID*: Unique identity number, unique at the national level for each citizen
3. *BankAccount_Num*: Unique account number at the bank. A student can have multiple accounts or joint accounts. This attribute stores the primary account number.
4. *Name*: Name of the student
5. *Hostel_Room*: Room number of the hostel

Which of the following options is **INCORRECT**?

(A) *BankAccount_Num* is a candidate key

(B) *Registration_Num* can be a primary key

(C) *UID* is a candidate key if all students are from the same country

(D) If S is a superkey such that $S \cap UID$ is NULL then $S \cup UID$ is also a superkey

Open solution

Q16. GATE CSE 2005

Which one of the following statements about normal forms is **FALSE?**

(A) $BCNF$ is stricter than $3NF$

(B) Lossless, dependency $-$ preserving decomposition into $3NF$ is always possible

(C) Lossless, dependency - preserving decomposition into $BCNF$ is always possible

(D) Any relation with two attributes is in $BCNF$

Open solution

Q17. GATE CSE 2005

A table has fields, $F1, F2, F3, F4, F5$, with the following functional dependencies:
$F1 \rightarrow F3$. $F2 \rightarrow F4$. $(F1\,.\,F2) \rightarrow F5$ in terms of Normalization, this table is in

(A) $1NF$ (B) $2NF$ (C) $3NF$ (D) None of these

Open solution

Q18. GATE CSE 2002

Relation R with an associated set of functional dependencies, F, is decomposed into $BCNF$. The redundancy (arising out of functional dependencies) in the resulting set of relations is

(A) Zero

(B) More than zero but less than that of an equivalent $3NF$ decomposition

(C) Proportional to the size of F^+

(D) Indeterminate

Open solution

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Q19. GATE CSE 2001

Consider a schema $R(A, B, C, D)$ and functional dependencies $A \rightarrow B$ and $C \rightarrow D$. Then the decomposition of R into $R_1\ (AB)$ and $R_2\ (CD)$ is

(A) Dependency preserving and lossless join

(B) Lossless join but not dependency preserving

(C) Dependency preserving but not lossless join

(D) Not dependency preserving and not lossless join

Open solution

Q20. GATE CSE 1999

Let $R = (A, B, C, D, E, F)$ be a relation scheme with the following dependencies: $C \rightarrow F$, $E \rightarrow A$, $EC \rightarrow D$, $A \rightarrow B$. Which of the following is a key for R?

(A) CD (B) EC (C) AE (D) AC

Open solution

Q21. GATE CSE 1998

Which normal form is considered adequate for normal relational database design?

(A) $2\ NF$ (B) $5\ NF$ (C) $4\ NF$ (D) $3\ NF$

Open solution

Q22. GATE CSE 1994

State True or False with reason. There is always a decomposition into Boyce-codd normal form $(BCNF)$ that is lossless and dependency preserving.

(A) TRUE (B) FALSE

Open solution

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Marks 2 34 questions

Q23. GATE CSE 2026 Set 2

Open solution

In the context of schema normalization in relational DBMS, consider a set F of functional dependencies. The set of all functional dependencies implied by F is called the closure of F. To compute the closure of F, Armstrong's Axioms can be applied. Consider X, Y, and Z as sets of attributes over a relational schema. The three rules of Armstrong's Axioms are described as follows.

Reflexivity: If $Y \subseteq X$, then $X \rightarrow Y$

Augmentation: If $X \rightarrow Y$, then $XZ \rightarrow YZ$ for any Z

Transitivity: If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$

The additional rule of Union is defined as follows.

Union: If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$

It can be proved that the additional rule of Union is also implied by the three rules of Armstrong's Axioms. Listed below are four combinations of these three rules. Which one of these combinations is both necessary and sufficient for the proof?

(A) Reflexivity, Augmentation, and Transitivity

(B) Reflexivity and Augmentation

(C) Transitivity

(D) Augmentation and Transitivity

Q24. GATE CSE 2026 Set 1

Open solution

Consider a relational database schema with a relation $R(A, B, C, D)$. If $\{A, B\}$ and $\{A, C\}$ are the only two candidate keys of the relation R, then the number of superkeys of relation R is ____. (answer in integer)

Q25. GATE CSE 2025 Set 2

Open solution

Consider the following relational schema along with all the functional dependencies that hold on them.

$R1(A, B, C, D, E) : \{D \rightarrow E, EA \rightarrow B, EB \rightarrow C\}$
$R2(A, B, C, D) : \{A \rightarrow D, A \rightarrow B, C \rightarrow A\}$

Which of the following statement(s) is/are TRUE?

(A) $R1$ is in 3NF

(B) $R2$ is in 3 NF

(C) $R1$ is NOT in 3 NF

(D) $R2$ is NOT in 3NF

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Q26. GATE CSE 2025 Set 1

Consider a relational schema team(name, city, owner), with functional dependencies \{name $\rightarrow$ city, name $\rightarrow$ owner}.

The relation team is decomposed into two relations, $t1$ (name, city) and $t2$ (name, owner). Which of the following statement(s) is/are TRUE?

(A) The relation team is NOT in BCNF.

(B) The relations $t1$ and $t2$ are in BCNF.

(C) The decomposition constitutes a lossless join.

(D) The relation team is NOT in 3NF.

Open solution

Q27. GATE CSE 2024 Set 2

A functional dependency $F : X \rightarrow Y$ is termed as a useful functional dependency if and only if it satisfies all the following three conditions:

- X is not the empty set.
- Y is not the empty set.
- Intersection of X and Y is the empty set.

For a relation R with 4 attributes, the total number of possible useful functional dependencies is _____

Open solution

Q28. GATE CSE 2024 Set 1

The symbol → indicates functional dependency in the context of a relational database. Which of the following options is/are TRUE?

(A) (X, Y) → (Z, W) implies X → (Z, W)

(B) (X, Y) → (Z, W) implies (X, Y) → Z

(C) ((X, Y) → Z and W → Y) implies (X, W) → Z

(D) (X → Y and Y → Z) implies X → Z

Open solution

Q29. GATE CSE 2021 Set 2

Suppose the following functional dependencies hold on a relation U with attributes P, Q, R, S, and T :

P → QR

RS → T

Which of the following functional dependencies can be inferred from the above functional dependencies?

(A) P → R (B) PS → T (C) R → T (D) PS → Q

Open solution

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Q30. GATE CSE 2021 Set 1

Open solution

Consider the relation R(P, Q, S, T, X, Y, Z, W) with the following functional dependencies.

PQ → X; P → YX; Q → Y; Y → ZW

Consider the decomposition of the relation R into the constituent relations according to the following two decomposition schemes.

D_1 : R = [(P, Q, S, T); (P, T, X); (Q, Y); (Y, Z, W)]

D_2 : R = [(P, Q, S); (T, X); (Q, Y); (Y, Z, W)]

Which one of the following options is correct?

(A) D_1 is a lossy decomposition, but D_2 is a lossless decomposition.

(B) Both D_1 and D_2 are lossless decompositions.

(C) Both D_1 and D_2 are lossy decompositions.

(D) D_1 is a lossless decomposition, but D_2 is a lossy decomposition.

Q31. GATE CSE 2020

Open solution

Consider a relational table R that is in 3NF, but not in BCNF. Which one of the following statements is TRUE?

(A) A cell in R holds a set instead of an atomic value.

(B) R has a nontrivial functional dependency X → A, where X is not a superkey and A is a non-prime attribute and X is not a proper subset of any key

(C) R has a nontrivial functional dependency X → A, where X is not a superkey and A is a non-prime attribute and X is a proper subset of some key.

(D) R has a nontrivial functional dependency X → A, where X is not a superkey and A is a prime attribute.

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Q32. GATE CSE 2019

Open solution

Let the set of functional dependencies

F = {QR → S, R → P, S → Q}

hold on a relation schema X = (PQRS). X is not in BCNF. Suppose X is decomposed into two schemas Y and Z, where Y = (PR) and Z = (QRS).

Consider the two statements given below.

I. Both Y and Z are in BCNF
II. Decomposition of X into Y and Z is dependency preserving and lossless

Which of the above statements is/are correct?

(A) I only **(B)** Neither I nor II **(C)** II only **(D)** Both I and II

Q33. GATE CSE 2018

Open solution

Consider the following four relational schemas. For each schema, all non-trivial functional dependencies are listed. The underlined attributes are the respective primary keys.

Schema I: Registration (rollno, courses)
Field 'courses' is a set-valued attribute containing the set of courses a student has registered for.
Non-trivial functional dependency:
rollno → courses

Schema II: Registration (rollno, courseid, email)
Non-trivial functional dependencies:
rollno, courseid → email
email → rollno

Schema III: Registration (rollno, courseid, marks, grade)
Non-trivial functional dependencies:
rollno, courseid → marks, grade
marks → grade

Schema IV: Registration (rollno, courseid, credit)
Non-trivial functional dependencies:
rollno, courseid → credit
courseid → credit

Which one of the relational schemas above is in 3NF but not in BCNF?

(A) Schema I **(B)** Schema II **(C)** Schema III **(D)** Schema IV

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Q34. GATE CSE 2014 Set 1

Open solution

Given the following two statements:
$S1$: Every table with two single-valued attributes is in $1NF, 2NF, 3NF$ and $BCNF$.
$S2$: $AB \rightarrow C,\ D \rightarrow E,\ E \rightarrow C$ is a minimal cover for the set of functional dependencies $AB \rightarrow C$, $D \rightarrow E,\ AB \rightarrow E,\ E \rightarrow C$.
Which one of the following is CORRECT?

(A) $S1$ is TRUE and $S2$ is FALSE
(B) Both $S1$ and $S2$ are TRUE.
(C) $S1$ is FALSE and $S2$ is TRUE.
(D) Both $S1$ and $S2$ are FALSE.

Q35. GATE CSE 2013

Open solution

Relation R has eight attributes ABCDEFGH. Fields of R contain only atomic values.

$F = \{CH \rightarrow G,\ A \rightarrow BC,\ B \rightarrow CFH,\ E \rightarrow A,\ F \rightarrow EG\}$ is a set of functional dependencies (FDs) so that F^+ is exactly the set of FDs that hold for R.

How many candidate keys does the relation R have?

(A) 3 (B) 4 (C) 5 (D) 6

Q36. GATE CSE 2013

Open solution

Relation R has eight attributes ABCDEFGH. Fields of R contain only atomic values.

$F = \{CH \rightarrow G,\ A \rightarrow BC,\ B \rightarrow CFH,\ E \rightarrow A,\ F \rightarrow EG\}$ is a set of functional dependencies (FDs) so that F^+ is exactly the set of FDs that hold for R.

The relation R is

(A) in $1NF$, but not in $2NF$.
(B) in $2NF$, but not in $3NF$.
(C) in $3NF$, but not in $BCNF$
(D) in $BCNF$.

Q37. GATE CSE 2009

Open solution

Consider the following relational schema:

Suppliers(sid: integer, sname: string, city: string, street: string)

Parts(pid: integer, pname: string, color: string)

Catalog(sid: integer, pid: integer, cost: real)

Assume that, in the suppliers relation above, each supplier and each street within a city has a unique name, and (sname, city) forms a candidate key. No other functional dependencies are implied other than those implied by primary and candidate keys. Which one of the following is TRUE about the above schema?

(A) The schema is in BCNF.
(B) The schema is in 3NF but not in BCNF.
(C) The schema is in 2NF but not in 3NF.
(D) The schema is not in 2NF.

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Q38. GATE CSE 2008

Open solution

Consider the following relational schemes for a library database:

Book (Title, Author, Catalog_no, Publisher, Year, Price)

Collection (Title, Author, Catalog_no)

with the following functional dependencies:

I. Title Author $\rightarrow$ Catalog_no

II. Catalog_no $\rightarrow$ Title Author Publisher Year

III. Publisher Title Year $\rightarrow$ Price

Assume {Author, Title} is the key for both schemes. Which of the following statements is true?

(A) Both Book and Collection are in $BCNF$

(B) Both Book and Collection are in $3\ NF$ only

(C) Book is in $2\ NF$ and Collection is in $3\ NF$

(D) Both Book and Collection are in $2\ NF$ only

Q39. GATE CSE 2007

Open solution

Which one of the following statements is **FALSE?**

(A) Any relation with two attributes is in $BCNF$.

(B) A relation in which every key has only one attribute is in $2NF$.

(C) A prime attribute can be transitively dependent on a key in a $3NF$ relation.

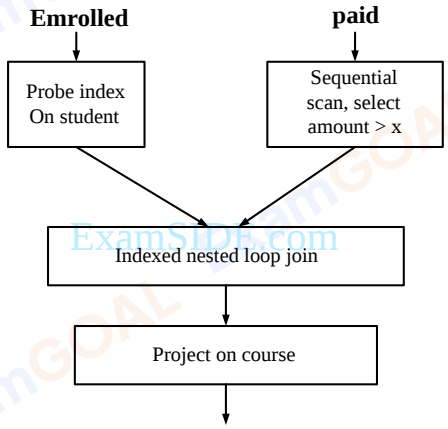
(D) A prime attribute can be transitively dependent on a key in a $BCNF$ relation.

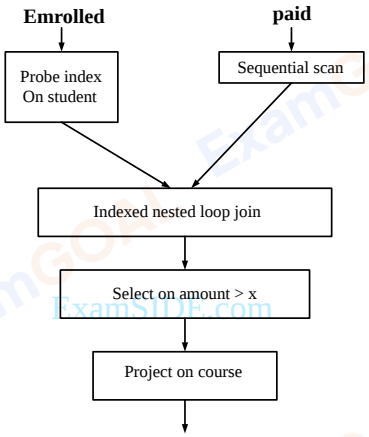
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Q40. GATE CSE 2006

Open solution

Consider the relation enrolled (student, course) in which (student, course) is the primary key, and the relation Paid (student, amount) where student is the primary key. Assume no null values and no foreign keys or integrity constraints. Assume that amounts $6000, 7000, 8000, 9000$ and 10000 were each paid by 20% of the students. Consider these query plans (Plan 1 on left, Plan 2 on right) to "List all courses taken by students who have paid more than x".





A disk seek takes $4ms$, disk data transfer bandwidth is $300MB/s$ and checking a tuple to see if amount is greater than x takes $10\mu s$. Which of the following statements is correct?

(A) Plan 1 and Plan 2 will not output identical row sets for all databases

(B) A course may be listed more than once in the output of plan 1 for some databases

(C) For $x = 5000$, Plan 1 executes faster than plan 2 for all databases

(D) For $x = 9000$, Plan 1 executes slower than plan 2 for all databases

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Q41. GATE CSE 2006

The following functional dependencies are given :
$AB \rightarrow CD,\ AF \rightarrow D,\ DE \rightarrow F,$
$C \rightarrow G. \quad F \rightarrow E. \quad G \rightarrow A.$
Which one of the following options is false?

(A) $\{CF\}^+ = \{ACDEFG\}$
(B) $\{BG\}^+ = \{ABCDG\}$
(C) $\{AF\}^+ = \{ACDEFG\}$
(D) $\{AB\}^+ = \{ABCDG\}$

Open solution

Q42. GATE CSE 2005

In a schema with attributes A, B, C, D, and E, following set of functional dependencies are given
$A \rightarrow B$
$A \rightarrow C$
$CD \rightarrow E$
$B \rightarrow D$
$E \rightarrow A$
Which of the following functional dependencies is NOT implied by the above set?

(A) $CD \rightarrow AC$
(B) $BD \rightarrow CD$
(C) $BC \rightarrow CD$
(D) $AC \rightarrow BC$

Open solution

Q43. GATE CSE 2005

Consider a relation scheme $R = (A,\ B,\ C,\ D,\ E,\ H)$ on which the following functional dependencies hold: $\{A \rightarrow B,\ BC \rightarrow D,\ E \rightarrow C,\ D \rightarrow A\}$. What are the candidate keys of R?

(A) AE, BE
(B) AE, BE, DE
(C) AEH, BEH, BCH
(D) AEH, BEH, DEH

Open solution

Q44. GATE CSE 2004

The relation scheme student Performance (Name, CourseNo, RollNo, Grade) has the following functional dependencies:
Name, courseNo $\rightarrow$ grade
RollNo, courseNo $\rightarrow$ grade
Name $\rightarrow$ rollNo
RollNo $\rightarrow$ name

The highest normal form of this relation scheme is

(A) $2\,NF$
(B) $3\,NF$
(C) $BCNF$
(D) $4\,NF$

Open solution

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Q45. GATE CSE 2004

A relation Empdt 1 is defined with attributes empcode (unique), name, street, city, state and pincode. For any pincode, there is only one city and state. Also, for any given street city and the state, there is just one pincode. In normalization terms, Empdt1 is a relation in

Open solution

(A) $1NF$ only

(B) $2NF$ and hence also in $1NF$

(C) $3NF$ and hence also in $2NF$ and $1NF$

(D) $BCNF$ and hence also in $3NF$, $2NF$ and $1NF$

Q46. GATE CSE 2003

Consider the following functional dependencies in a database.

$$Date\ of\ Birth \rightarrow Age$$
$$Age \rightarrow Eligibility$$
$$Name \rightarrow Roll_number$$
$$Roll_number \rightarrow Name$$
$$Course_number \rightarrow Course_name$$
$$Course_number \rightarrow Instructor$$
$$(Roll_Number, Course_number) \rightarrow Grade$$

The relation (Roll_number, Name, Date_of_Birth, Age) is

Open solution

(A) In $2\ NF$ but not in $3\ NF$

(B) In $3\ NF$ but not in $BCNF$

(C) In $BCNF$

(D) None of the above

Q47. GATE CSE 2002

Relation R is decomposed using a set of functional dependencies, F, and relation S is decomposed using another set of functional dependencies, G. One decomposition is definitely $BCNF$, the other is definitely. $3NF$, but it is not known which is which. To make a guaranteed identification, which one of the following tests should be used on the decompositions? (Assume that the closures of F and G are available).

Open solution

(A) Dependency-preservation

(B) Lossless-join

(C) $BCNF$ definition

(D) $3\ NF$ definition

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Q48. GATE CSE 2002

Open solution

From the following instance of a relation schema $R(A, B, C)$, we can conclude that:

A	B	C
1	1	1
1	1	0
2	3	2
2	3	2

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(A) A functionally determines B and B functionally determines C

(B) A functionally determines B and B does not functionally determines C

(C) B does not functionally determines C

(D) A does not functionally determine B and B does not functionally determine C

Q49. GATE CSE 2001

Open solution

$R(A, B, C, D)$ is a relation. Which of the following does not have a lossless-join, dependency preserving $BCNF$ decomposition?

(A) $A \rightarrow B$, $B \rightarrow CD$

(B) $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$

(C) $AB \rightarrow C$, $C \rightarrow AD$

(D) $A \rightarrow BCD$

Q50. GATE CSE 2000

Open solution

Given the following relation instance

X	Y	Z
1	4	2
1	5	3
1	6	3
3	2	2

Which of the following functional dependencies are satisfied by the instance?

(A) $XY \rightarrow Z$ and $Z \rightarrow Y$

(B) $YZ \rightarrow X$ and $Y \rightarrow Z$

(C) $YZ \rightarrow X$ and $X \rightarrow Z$

(D) $XZ \rightarrow Y$ and $Y \rightarrow X$

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Q51. GATE CSE 1999

Consider the schema $R = (S\ T\ U\ V)$ and the dependencies $S \to T,\ T \to U,\ U \to V$ and $V \to S$ let $R = (R1$ and $R2)$ be a decomposition such that $R1 \cap R2 \neq \phi$. The decomposition is:

(A) Not in $2NF$

(B) In $2NF$ but not $3NF$

(C) In $3NF$ but not in $2NF$

(D) In both $2NF$ and $3NF$

Open solution

Q52. GATE CSE 1998

Consider the following database relations containing the attributes
Book–id
Subject–Category–of–book
Name–of–Author
Nationality–of–Author
With book–id as the primary key.
(a) What is the highest normal form satisfied by this relation?
(b) Suppose the attributes Book–title and Author–address are added to the relation, and the primary key is changed to {Name–of–Author, Book–title}, what will be the highest normal form satisfied by the relation?

Open solution

Q53. GATE CSE 1998

Consider the following database relations containing the attributes
Book–id
Subject–Category–of–book
Name–of–Author
Nationality–of–Author
With book–id as the primary key.
(a) What is the highest normal form satisfied by this relation?
(b) Suppose the attributes Book–title and Author–address are added to the relation, and the primary key is changed to {Name–of–Author, Book–title}, what will be the highest normal form satisfied by the relation?

Open solution

Q54. GATE CSE 1997

For a database relation $R(a, b, c, d)$, where the domains of a, b, c, d include only atomic values, only the following functional dependencies and those that can be inferred from them hold:
$a \to c \qquad b \to d$
This relation is

(A) In $1\ NF$ but not in $2\ NF$

(B) In $2\ NF$ but not in $3\ NF$

(C) In $3\ NF$

(D) None of the above

Open solution

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Q55. GATE CSE 1997

Open solution

Let $R(a, b, c)$ and $S(d, e, f)$ be two relations in which d is the foreign key of S that refers to the primary key of R. Consider the following four operations R and S

(a) Insert into R (b) Insert into S

(c) Delete from R (d) Delete from $$S$

Which of the following statements is true about the referential integrity constraint above?

(A) None of (a), (b), (c) or (d) can cause its violation

(B) All of (a), (b), (c) and (d) can cause its violation

(C) Both (a) and (d) can cause its violation

(D) Both (b) and (c) can cause its violation

Q56. GATE CSE 1995

Open solution

(a) Consider the relation scheme $R(A, B, C)$ with the following functional dependencies:

$A, B \rightarrow C$

$C \rightarrow A$

Show that the scheme R is the Third Normal Form $(3NF)$ but not in Boyce-Code Normal Form $(BCNF)$.

(b) Determine the minimal keys of relation R.

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Answer Key

Marks 1

1. **C, D**	2. **A, B, C**	3. **A, C**	4. **A**	5. **8**
6. **C**	7. **A**	8. **B**	9. **C**	10. **B**
11. **19**	12. **8**	13. **B**	14. **C**	15. **A**
16. **C**	17. **A**	18. **A**	19. **C**	20. **B**
21. **D**	22. **B**			

Marks 2

23. **D**	24. **(6)**	25. **C, D**	26. **B, C**	27. **50**
28. **B, C, D**	29. **A, B, D**	30. **D**	31. **D**	32. **C**
33. **B**	34. **A**	35. **B**	36. **A**	37. **A**
38. **C**	39. **D**	40. **C**	41. **C**	42. **B**
43. **D**	44. **B**	45. **B**	46. **D**	47. **C**
48. **C**	49. **C**	50. **B**	51. **D**	52. **(a)** $2NF$ **(b)** $3NF$
53. **(a)** $2NF$ **(b)** $3NF$	54. **A**	55. **D**	56. **Solve it.**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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